

Assessment 2:

Android Application Design Proposal and Skeleton Prototype (20%)

Group Assignment (Groups of 4)

Group Assignment (Groups of 4) All group members should be from the same lab

Submission: A zip file via Moodle by **Friday, 24 April 2026 at 11:55 PM (Week 7)**

This assessment is **worth 20%** of the total mark for this unit.

Group Requirements

- Each group must have **FOUR members** (unless approved by unit CE/UC otherwise)
- All group members should be **from the same lab section, with no exception permitted.**

Assessment Objective

The purpose of this assignment is to **propose the design specifications** of a mobile distributed application and to **implement the app's screens using Jetpack Compose** in Android Studio. You will then develop a fully functional app based on this proposal in Assessment 4 (Android App Development and Demo). Therefore, the **group members must remain the same for both Assessment 2 and Assessment 4.**

This assessment supports unit learning outcomes 1 and 2:

- **LO1:** Identify and describe different approaches and methods for building distributed and mobile computing systems.
- **LO2:** evaluate several models and approaches and select a suitable mobile computing solution for a particular case.

Overview and Deliverables

For this assessment, your group will propose the design and implementation of a mobile distributed application in Android Studio (using an emulator). The app will interact with public web APIs such as a weather API, and cloud-based services like Google Firebase (as the backend).

In Assessment 4, your group will fully develop this app in Android Studio based on your proposal. This assignment is an opportunity to apply your creative and technical skills to design an innovative mobile app that addresses real-world problems or provides a benefit to society.

This assessment requires you to **write a report** describing the details of your app based on the provided instructions and to **create a prototype**. Here, **the prototype refers to the implementation of the app's screens using Jetpack Compose** in Android Studio.

This assignment requires you to **submit a zip file to Moodle** containing the following:

1. **A project design proposal report using the template provided in Appendix A**
 - The word limit of the proposal is **1200 words (including acknowledgement of using AI, and a detailed history of AI usage that is not included in the word count)**.
2. **A task allocation form (Appendix B).**
3. **Your entire Android project (zipped)** including Composable functions (please zip it before packaging): your group is required to implement a **skeleton prototype**, which provides a scaffolded version of the application, including core screen structures, navigation flow, and essential boilerplate code. The skeleton prototype is not expected to be fully functional, but it should demonstrate the overall architecture and layout of the app, serving as a foundation for future development.
4. **Optional files to include:** any original design graphics, images, screenshots, prototype diagram etc. of those embedded in your report to improve readability.

To complete the assignment, please follow these steps:

The Design Proposal - Key steps (following the template in Appendix A)

Step 1. Introduction

1.1. Application domain and SDG goals

As a group, you will first **select an application domain** (for example, health, education, smart energy, green environment, wildlife saving, smart gardening, among many others), and then **identify its goal based on the 17 UN Sustainable Development Goals (SDGs)** (refer to

<https://www.un.org/sustainabledevelopment/sustainable-development-goals/>)

Below are some examples of application domains and the SDG goals to inspire you, but your app and its functionalities are not limited to these:

- Wildlife Saving or Care App (SDG 15) – animal facts, saving strategies, reporting options, rescue hotline, map of wildlife-friendly zones

- Food Sharing and Waste Reduction App (SDG 2) – share leftover meals, find donation spots, food expiry reminders
- Affordable Health & Wellness Tracker (SDG 3) – mood diary, home exercises, reminders for medication.
- Women’s Safety and Empowerment App (SDG 5) – SOS button, location sharing, self-defence tips
- Clean Water Access & Usage Monitor (SDG 6) – daily water-saving tips, water source maps, usage calculator
- Energy Saver & Carbon Footprint Tracker (SDG 7) – eco-friendly tips, power usage tracker, CO₂ reduction goals
- Eco-Friendly Travel Guide (SDG 11) – bus/train schedules, bike route maps, carpool finder
- Waste Sorting and Recycling Helper (SDG 12) – scan an item to know how to recycle, recycling centre map

In this section, you need to **briefly describe the application domain and explain why it aligns with one of the SDG goals**.

Note: application domains with low impact, such as task or event trackers or managers, are **not acceptable**.

1.2 Impact and value

The proposed mobile app must demonstrate a clear **impact** on **improving people’s lives, communities, society, or the world**. This impact can be achieved by addressing a real problem, filling an existing gap, or enhancing the way people currently perform certain tasks. For example, the app may automate or simplify processes, increase accessibility, raise awareness, or encourage positive behavioural change.

In addition to impact, the app **must provide value** to its users. Value can come in many forms, such as:

- Practical value – making a task faster, easier, or more cost-effective
- Educational value – helping users gain knowledge or new skills
- Health and well-being value – improving mental or physical health
- Social value – fostering connections, inclusion, or equality
- Environmental value – supporting sustainability, conservation, or climate action
- Economic value – creating opportunities or supporting livelihoods

In this section, **briefly describe the expected impact and value** of the proposed app.

1.3 User group

In this section, you will describe the intended user group, including who they are (e.g., age, gender, profession) and their specific needs or abilities (if any). Explain **who benefits from the app’s impact**. This could be specific individuals, communities, or groups. Include relevant details such as their level of technological familiarity, accessibility requirements, and how they are expected to interact with the app.

The user group must be described in a clear, written paragraph. **Personas are not allowed**.

1.4 Context-awareness

The proposed mobile app **must be context-aware**, meaning it should **collect contextual information** and **use it for context-aware computing**. The concepts of context and context-aware computing will be **explained during the Week 4 seminar**.

Since this unit uses a virtual device, you will need to **simulate real sensors** by utilising **publicly available sensory datasets** (e.g., smart meter readings, accelerometer data, steps taken, heart rate, or glucose level)

and by writing a **class that continuously generates sensor data** to mimic real sensor behaviour. This can be easily achieved by reading from a data file and publishing the values at short intervals.

One of the publicly available datasets you may use is from Kaggle: <https://www.kaggle.com/datasets>

Additionally, you will **create a class that uses this sensor data**, along with other sensory and non-sensory data (**multiple contextual information**) to dynamically **adjust the parameters or settings** of other parts of the app or to **enable personalisation** based on the current context and user preferences.

You can also **incorporate a mobile AI model** into the project to perform predictions or detections using this data (**as an Advanced Feature, see Step 5**).

Step 2. Android components, App functionalities, and Screens

2.1. Android components

The proposed app **must incorporate the following Android components** covered in this unit:

1. **Room database**
2. **Navigation (Navigation Rail or Bottom Bar Navigation (avoid using both), Drawer is optional if combined with others)**
3. **Retrofit**
4. **DatePicker**
5. **Expanded Dropdown Menu**
6. **LazyColumn**

In this section of the report, you must list all six of these components and include a brief sentence describing their purpose in your app. For example, for Room, specify what data it will store, and for Retrofit, indicate which API it will interact with.

In addition to these basic components, you **must include 3-4 additional Android components** such as Checkbox, TimePicker, Switch, RadioButton, Progress indicators, or any other useful features relevant to your app.

You may use a table to present this section.

2.2 App Functionalities

Based on Section 1 (Introduction) and **aligned with the prototype** (screens) you created, this section should list the app's main functionalities. The app **must include the following core functionalities**:

1. To allow existing users to Login/Sign-in
2. To allow new users to Register/Sign-up
3. To collect data from users using mobile forms (e.g., Profile or Preferences)
4. Displaying recorded data or items, with options for users to select, add, delete, or edit them
5. Search functionality to allow users to find items or information (stored in Room, Google Firebase/Firestore, or retrieved from public APIs)
6. Data visualisation using charts/graphs to present data stored in Room or Google Firebase/Firestore, OR Map integration for location awareness or tracking

In addition to these six functionalities, you **must include 1-2 additional features** specific to your app that **demonstrate creativity and complexity**.

As the first additional feature, for functionalities 1 and 2, you can propose using **Google Authentication to manage user login** and **Google Firebase/Firestore to store non-sensitive data** that can be shared. As a

second additional feature, you can propose a real-time messaging that can be later implemented as a chat system using Firebase or Firestore to send and receive messages in real time between two running emulators. **You may use a table to present this section.**

2.3 Screens and User Interface

In this section, you need to list the screens your app will have and **what each screen's main purpose is**. For instance, a pain diary app might have a login screen, signup screen, home screen, pain entry screen, history screen, report screen, and search for a doctor screen.

Screens must match the proposed functionalities.

You may use a table to present this section.

Step 3. The UI Design and Android UI Skeleton Prototype

Based on the functionalities and screens you proposed, you will create **a prototype of your app (only the UI screens, without any functional code)** in Android using **Jetpack Compose** (Composable functions).

In this section, you need to provide **ONE diagram of your prototype** that includes **the screenshots of all the main screens** created using Jetpack Compose, along **with arrows showing the navigation flow** between them. Each screen should have a **meaningful title** where applicable. You can create this diagram in **Word, PowerPoint, or a similar tool** by inserting the app screen screenshots and adding arrows between them to illustrate the navigation flow. You **must run the emulator in Android Studio** to capture a screenshot of each app screen.

In addition to the single diagram, you must also include up to **FOUR screenshots of individual app screens per page**, each with a brief description, so we can assess them in detail.

You must **include the Android project code** (the composable functions used to create the screens) in your **submission to Moodle**.

Skeleton Prototype: Your group/team is required to implement a skeleton prototype, which provides a scaffolded version of the application, including core screen structures, navigation flow, and essential boilerplate code. The skeleton prototype is not expected to be fully functional, but it should demonstrate the overall architecture and layout of the app, serving as a foundation for future development.

Step 4 – Design Guidelines

In designing the screens, you need to consider **the design guidelines** that we cover in the seminar (Week 6).

- You need to list **TWO password guidelines** for the login/sign-up screen.
- For the form screen (where you collect user data), you need to list **THREE specific and clear form design guidelines** that can be easily justified. Avoid broad or general guidelines. These requirements will be **explained during the Week 6 seminar**.

Clearly **indicate where these FIVE guidelines are applied** by **referring to the screen title and specific UI elements**.

Step 5 – Advanced Features

As part of this proposal, you also need to select and propose **TWO advanced features** from the list below that will be implemented in your proposed mobile app in Assessment 4.

Part 1 - In your description, clearly explain how each selected feature will **benefit and enhance your app**. This section will be assessed based on the **relevance, usefulness, and meaningfulness** of the chosen features.

Part 2- You need to **briefly describe how you will learn and implement these features** in your app by providing at least **THREE online sources per feature** (e.g., online tutorials, libraries, GitHub pages, or sample source code). This means **a total of six** well-researched sources.

Two advanced features must be selected from the following list:

1. **Android WorkManager** to run a job in the background continuously, e.g., collecting sensory data and writing it to a file or running an AI/ML model for activity recognition using accelerometer data.
2. **To schedule reminders** within the app using **AlarmManager**, triggering a notification or in-app alert at the specified time. For example, reminding the user to take medication at a certain time.
3. **Incorporating Google Calendar API via the SDK** (NOT just simple intents to open the calendar app). This would involve setting up Google Cloud Console credentials, using OAuth 2.0 authentication, integrating the Google Calendar SDK (REST API), and handling API calls.
4. **Running a mobile version of an AI model** (e.g., using the TensorFlow Lite library) in the app for detection or prediction.
5. **Google's ML Kit for text recognition from images**.

To include any other advanced feature, you need **permission from your tutor**.

Step 6. Task Allocation

You also need to **complete the Task Allocation form (Appendix B)**, where you will assign the tasks among the group members **for Assessment 4** (NOT for this assessment) based on this proposal.

File Submission

- **File Name Format:** Use this format: **A2-Lab0X-GNo.zip** (e.g., **A2-Lab01-G01.zip**).
- **Submission: Only ONE group member** should submit a zip file to Moodle, including:
 - **the report (PDF file, including acknowledgement of using AI, and a detailed history of AI usage that is not included in the word count),**
 - **the Task Allocation Form, and**
 - **zipped Codes: Your complete Android project must be packaged as a .zip file first, next to the above TWO documents.**
 - **Optional files to include:** any original design graphics, images, screenshots, prototype diagram etc. of those embedded in your report to improve readability.
 - Make sure the **submitted ZIP file can be opened without any errors**.
- **Deadline:** Friday, 24 April 2026 at 11:55 PM (Week 7).

(Moodle may show a late submission notification to other members; please ignore this, as we are aware of it. The notification will disappear after marks are uploaded.)

Policies and Penalties

Late Submissions

There is a **5% penalty per day** for late submissions.

Reference:

<https://www.monash.edu/learning-teaching/teachhq/Assessment/marking-and-grading/how-to/calculating-late-penalties>

Academic Integrity and Generative AI Use

Academic Integrity Rules

Generative AI tools can be used for **one or more parts** of the assessment task (**not the entire task**)

Represent a sincere **demonstration of human efforts and subject knowledge**

You must **acknowledge** the use of Generative AI with descriptions, or any other tool (including templates) used to create slides

Acknowledgement Requirements:

Please note: **Generative AI tools are not restricted** for this assessment task **BUT** a combination of the following should be provided to acknowledge the use of generative AI in academic work:

- **Written acknowledgment of the use of generative artificial intelligence** and its extent;
- **Descriptions** of how the information was generated (**including the prompts used**); and
- Summarise **how you have altered, adopted, or built on the AI output**.

When marking assessments, if unauthorised use of generative AI is suspected or **unacknowledged work is discovered**, it will be managed **as an allegation of academic misconduct** in accordance with existing disciplinary processes.

- *Guidance on Acknowledging AI Use:*
<https://www.monash.edu/learnhq/build-digital-capabilities/create-online/acknowledging-the-use-of-generative-artificial-intelligence>

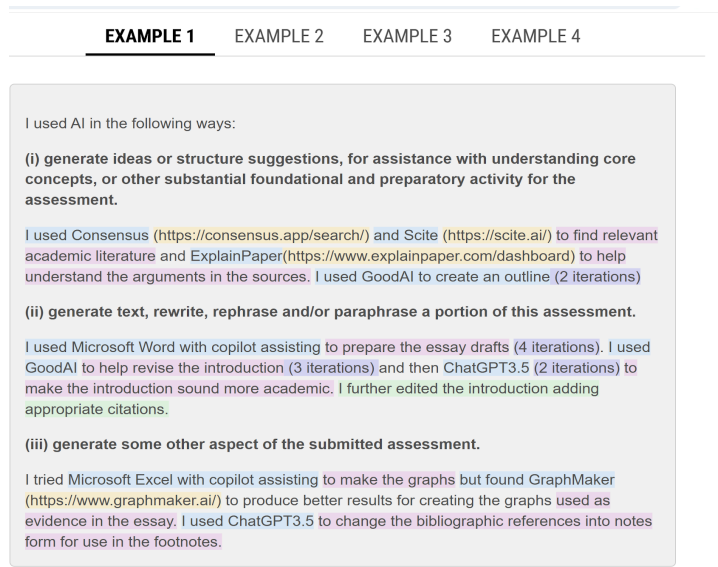
Example Acknowledgment (from Monash website):

Prompt: Was Sir John Monash a good man?

"Sir John Monash is widely regarded as one of Australia's greatest military leaders" (OpenAI, 2023)

Reference

OpenAI. (2023). *ChatGPT* (Jan 9 version) [Large language model] <http://chat.openai.com/chat>



Refer to the seminar slides (**How to use Gen AI in FIT5046 assessments?**) under Week 4 on Moodle.

Appendix A

The project proposal Report template

The proposal must have all the following headings. Each section is explained in the assignment under the Steps.

1 Introduction

- 1.1. Application domain and SDG goals
- 1.2. Impact and value
- 1.3. User group
- 1.4. Context-awareness

2 Android components, App functionalities, and Screens

- 2.1 Android components
- 2.2 App functionalities
- 2.3 Screens

3 The UI Design and Android Prototype (Jetpack Compose)

- **ONE diagram** of your prototype that includes the **screenshots** of the screens and the navigation flow between them **using arrows**.
- **4 screenshots of the app screens per page** and a brief description for each screen

4 Design Guidelines

- **TWO** password guidelines
- **THREE** form design guidelines

5 Advanced features

- Describe **TWO** selected advanced features and their purposes, and justify how they will add value to your proposed app
- Provide a total of **six** well-researched sources (three sources per feature)

6 References (the publications and website links based on the APA style)

Correct use of **in-text citation** for citing from the publications and website links you mentioned in the proposal. Provide a **Reference List** at the end of the report with at least 5 references using the APA style for your references, including URLs.

Monash University Library. (2020). Citing and Referencing: APA.

<https://guides.lib.monash.edu/citing-referencing/apa7th>

Appendix B

FIT5046 Task Allocation Form

Please provide the names of tasks and the name of the group member(s) who did the task in the table below. This form will represent your **proposed task allocation for Assessment 4**. You can make changes to it when you resubmit it for Assessment 4.

App Name:	Group Number:
Task Name	Member(s) responsible for
<i>The functionalities, screens, tasks, or components that each group member will develop later in Assignment 4 based on this proposal</i>	

Mark Deduction Guidance (Non-Compliance Penalties)

The following penalties will be applied for non-compliance with the assessment requirements, in addition to deductions based on the grading rubric:

Non-Compliance Issue	Mark Deduction (Applied to the Total Mark of A2)	Rationale
Submission Format Mismatch (e.g., incorrect file type and/or incorrect file naming convention)	-2%-5%	Failure to follow clear submission instructions, impacting marking efficiency.
Missing Required File(s) from the Original Submission	-10% per missing/incorrect file as required	The teaching team will follow up on any missing or incorrect file(s) to enable marking; however, failure to submit any required file in the original submission will result in a 10% mark deduction.
Missing/Incorrect In-Text Citations and Full References	-1%-5% per missing/incorrect citation	Failure to demonstrate research and referencing skills, and/or failure to adhere to APA style and documentation requirements.
Late Submission	-5% penalty per day	As per Monash late submission policy.
Academic Misconduct /Unacknowledged AI Use	In extreme cases, this may result in a mark of 0% for the entire assessment.	Will be managed as an allegation of academic misconduct per University policy.